HARD · GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

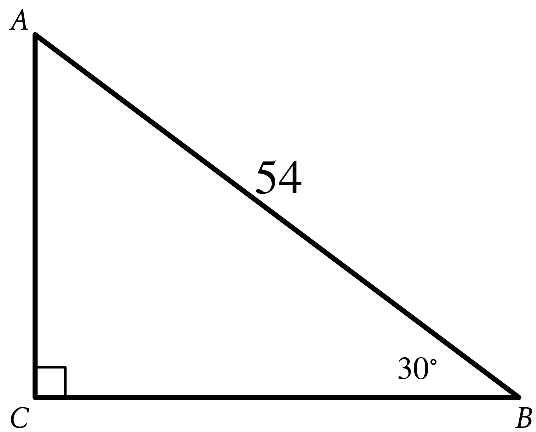
12

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- The length of side upper A upper B is 54.
- The measure of angle upper B is 30°.
- Angle upper C is a right angle.
- A note indicates the figure is not drawn to scale.

Right triangle ABC is shown. What is the value of $\tan A$?

A. $\frac{\sqrt{3}}{54}$

B. $\frac{1}{\sqrt{3}}$

C. $\sqrt{3}$

D. $27\sqrt{3}$

Q2 GEOMETRY AND TRIGONOMETRY

The circumference of the base of a right circular cylinder is 20π meters, and the height of the cylinder is 6 meters. What is the volume, in cubic meters, of the cylinder?

A. 60π

B. 120π

C. 600π

D. $2{,}400\pi$

Q3 GRID-IN GEOMETRY AND TRIGONOMETRY

The graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy-plane is a circle. What is the length of the circle's radius?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN GEOMETRY AND TRIGONOMETRY

The perimeter of an equilateral triangle is 852 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GEOMETRY AND TRIGONOMETRY

Points Q and R lie on a circle with center P. The radius of this circle is 9 inches. Triangle PQR has a perimeter of 31 inches. What is the length, in inches, of $\overline{QR}$?

A. $13\sqrt{2}$

B. 13

C. $9\sqrt{2}$

D. 9

Q6

GEOMETRY AND TRIGONOMETRY

In triangle XYZ, the measure of angle X is $90°$. Point W lies on segment YZ, and segment WX is perpendicular to segment YZ. The length of segment WY is 572, and the length of segment WX is 429. What is the value of $\tan Z$?

A. $\frac{3}{5}$

B. $\frac{3}{4}$

C. $\frac{4}{5}$

D. $\frac{4}{3}$

Q7

GEOMETRY AND TRIGONOMETRY

A circle has center O, and points A and B lie on the circle. The measure of arc AB is $45°$ and the length of arc AB is 3 inches. What is the circumference, in inches, of the circle?

A. 3

B. 6

C. 9

D. 24

Q8

GRID-IN GEOMETRY AND TRIGONOMETRY

The length of a rectangle's diagonal is $3\sqrt{17}$, and the length of the rectangle's shorter side is 3. What is the length of the rectangle's longer side?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GEOMETRY AND TRIGONOMETRY

$$x^2 + 14x + y^2 = 6y + 109$$

In the xy-plane, the graph of the given equation is a circle. What is the length of the circle's radius?

A. $\sqrt{109}$

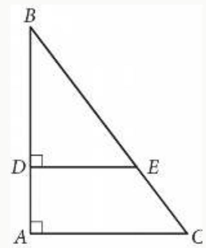
B. $\sqrt{149}$

C. $\sqrt{167}$

D. $\sqrt{341}$

Q10

GRID-IN GEOMETRY AND TRIGONOMETRY



In the figure above, $\tan B = \frac{3}{4}$. If $BC = 15$ and $DA = 4$, what is the length of $\overline{DE}$?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.